Lecture Scribe Preparation

Tutorial_1_Solution.pdf
PDF

You are required to prepare a formal lecture scribe from the attached PDF:

Tutorial-1: Solution
CSE 400: Fundamentals of Probability in Computing
School of Engineering and Applied Science (SEAS), Ahmedabad University
February 13, 2026

This lecture contains both practice questions and their complete solutions.
The final output must be a single unified formal scribe containing both questions and answers together.

Purpose

Prepare a formal lecture scribe using the provided lecture slides. The scribe must serve as exam reference material and contain definitions, notations, assumptions, preliminaries, main results, proofs, conclusions, and worked examples exactly as presented in the lecture slides. Use only the lecture content. Do not add explanations, interpretations, assumptions, or examples not present in the lecture slides.

REQUIRED STRUCTURE (DO NOT CHANGE ORDER)

Definitions and Notations

Assumptions and Preliminaries

Main Result

Proof

Conclusion

Working Examples

STRICT REQUIREMENTS (ALL MUST BE FOLLOWED)

Use EXACTLY the section order listed above.

Use ONLY the content present in the attached PDF.

Include BOTH:

All questions (Q1–Q10)

Their complete solutions exactly as written

Under Main Result, restate each question exactly as presented.

Under Proof, include the complete derivations and solutions exactly as shown in the PDF.

Under Working Examples, include:

Final computed answers

Final probability expressions

Final simplified inequalities

Final conclusions for each problem

Preserve exactly:

All mathematical expressions

All factorials and multinomial forms

All Poisson PMF formulas

All binomial expressions

All Gaussian PDF expressions

All Q-function expressions

All inequalities (⩾, ⩽, etc.)

All parameters (λ, θ, p, α, etc.)

All summations and limits

All conditional probability formulas

All step-function definitions

Maintain the same symbolic notation as the PDF (do not rename variables).

Do NOT:

Add explanations

Add intuition

Add textbook theory

Add extra steps not written in the PDF

Simplify wording

Omit intermediate steps

Modify phrasing

If any section (e.g., Conclusion) is not explicitly written in the lecture, leave that section blank.

Maintain a formal academic tone suitable for exam reference material.

Special Formatting Instruction

Since this is a solution lecture, structure each problem as:

Question (under Main Result)

Full derivation (under Proof)

Final answer clearly shown

No commentary outside what appears in the PDF

All content must appear in one continuous structured scribe.

ABSOLUTE RULE

This is an exam-reference document.

Do NOT expand.
Do NOT interpret.
Do NOT summarize.
Do NOT explain in your own words.
Use only the lecture content exactly as written.

Below is the formal lecture scribe prepared strictly from the provided PDF.

All content is taken exactly from:

Tutorial-1: Solution
CSE 400: Fundamentals of Probability in Computing
School of Engineering and Applied Science (SEAS), Ahmedabad University
February 13, 2026

Source:

Tutorial-1: Solution

CSE 400: Fundamentals of Probability in Computing
School of Engineering and Applied Science (SEAS), Ahmedabad University
February 13, 2026

Definitions and Notations

Poisson probability mass function:

$$P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}$$

Gaussian PDF:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

Gaussian relations:

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right), \quad Pr(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

Cumulative distribution function:

$$F(a) = \sum_{\text{all } x \leqslant a} p(x)$$

If X is discrete with possible values $x_1 < x_2 < x_3 < \cdots$,
then F is a step function.

Assumptions and Preliminaries

- Multinomial counting formula.
- Uniform distribution over an interval.
- Complement rule.
- De Morgan's law.
- Conditional probability formula.
- Binomial distribution.
- Poisson distribution.
- Gaussian distribution.
- Independence of events.
- Jury decision conditioning on guilt with probability α.

Main Result

Q1

You are organizing a Food Festival at Ahmedabad University featuring 20 delicious dishes from four different cuisines: Italian, Indian, Chinese, and Mexican. To create the perfect menu, you need to divide these 20 unique dishes into four equally delicious platters, each with exactly five dishes, ensuring that the order of the platters does not matter.

(a) Calculate the total number of ways you can divide these 20 different dishes into four groups of five dishes each.

(b) If there are five dishes from each cuisine, what is the probability that all four platters have dishes of the same cuisine?

Q2

Buses arrive at a specified stop at 15-minute intervals starting at 7 A.M. A passenger arrives at the stop at a time that is uniformly distributed between 7:00 and 7:30. Find the probability that the passenger waits:

(a) less than 5 minutes for a bus,
(b) more than 10 minutes for a bus.

Q3

Suppose that an employee arrives late 15% of the time, leaves early 25% of the time, and both arrives late and leaves early 8% of the time. What is the probability that on a given day the employee arrives early given that they leave late?

Q4

The expected number of typographical errors on a page is 0.2. What is the probability that the next page contains

(a) 0 errors?
(b) 2 or more errors?

Q5

The monthly worldwide average number of airplane crashes is 3.5. What is the probability that there will be

(a) at least 2 accidents?
(b) at most 1 accident?

Q6

Suppose that the number of accidents occurring on a highway each day is a Poisson random variable with parameter $\lambda = 3$.

(a) Find $P\{X \geqslant 3\}$
(b) Find $P\{X \geqslant 3 \mid X \geqslant 1\}$

Q7

A random variable X has a PDF given by:

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

Find each of the following probabilities and express in terms of Q-functions:

1. $Pr(X \leqslant 0)$
2. $Pr(X > 4)$
3. $Pr(|X+3| < 2)$
4. $Pr(|X-2| > 1)$

Q8

Consider a jury trial in which it takes 8 of the 12 jurors to convict the defendant. If we assume that jurors act independently and that, whether or not the defendant is guilty, each makes the right decision with probability θ, what is the probability that the jury renders a correct decision?

Q9

The probability mass function of a random variable X is given by

$$p(i) = \frac{c\lambda^i}{i!}, \quad i = 0, 1, 2, \ldots$$

Find:

(a) $P\{X = 0\}$
(b) $P\{X > 2\}$

Q10

A communication system consists of n components, each of which will independently function with probability p. The total system will be able to operate effectively if at least one-half of its components function.

(a) For what values of p is a 5-component system more likely to operate effectively than a 3-component system?

(b) In general, when is a $(2k + 1)$-component system better than a $(2k - 1)$-component system?

Proof

(All derivations reproduced exactly from the lecture PDF)

Q1

$$\text{Total ways} = \frac{20!}{(5!)^4 4!}$$

$$P = \frac{1}{\frac{20!}{(5!)^4 4!}} = \frac{(5!)^4 4!}{20!}$$

Q2

$$P(\text{wait} < 5) = \frac{10}{30} = \frac{1}{3}$$

$$P(\text{wait} > 10) = \frac{10}{30} = \frac{1}{3}$$

Q3

$$P(L) = 0.15, \quad P(E) = 0.25, \quad P(L \cap E) = 0.08$$

$$P(E^c) = 0.75$$

$$P(L \cup E) = 0.32$$

$$P(L^c \cap E^c) = 0.68$$

$$P(L^c \mid E^c) = \frac{0.68}{0.75} \approx 0.907$$

Q4

$$P(X = 0) = e^{-0.2} = 0.8187$$

$$P(X \geqslant 2) = 1 - 0.8187 - 0.1637 = 0.0176 \approx 0.0175$$

Q5

$$P(X \geqslant 2) = 0.8641$$

$$P(X \leqslant 1) = 0.1359$$

Q6

$$P(X \geqslant 3) = 0.5768$$

$$P(X \geqslant 3 \mid X \geqslant 1) = \frac{0.5768}{0.9502} \approx 0.607$$

Q7

$$Pr(X \leqslant 0) = 1 - Q(\frac{3}{2})$$

$$Pr(X > 4) = Q\left(\frac{7}{2}\right)$$

$$Pr(|X + 3| < 2) = 1 - 2Q(1)$$

$$Pr(|X - 2| > 1) = 1 - Q(2) + Q(3)$$

Q8

$$\alpha \sum_{i=8}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i} + (1-\alpha) \sum_{i=5}^{12} \binom{12}{i} \theta^i (1-\theta)^{12-i}$$

Q9

$$c = e^{-\lambda}$$

$$P(X = 0) = e^{-\lambda}$$

$$P(X > 2) = 1 - e^{-\lambda} - \lambda e^{-\lambda} - \frac{\lambda^2 e^{-\lambda}}{2}$$

Step function example:

$$F(a) = \begin{cases} 0, & a < 1 \\ \frac{1}{4}, & 1 \leqslant a < 2 \\ \frac{3}{4}, & 2 \leqslant a < 3 \\ \frac{7}{8}, & 3 \leqslant a < 4 \\ 1, & 4 \leqslant a \end{cases}$$

Q10

(a)

$$3(p-1)^2(2p-1) > 0$$

$$p > \frac{1}{2}$$

(b)

$$P_{2k+1} - P_{2k-1} = \binom{2k-1}{k} p^k (1-p)^k (2p-1)$$

$$> 0 \;\Leftrightarrow\; p > \frac{1}{2}$$

Conclusion

Working Examples

Q1:

$$\frac{20!}{(5!)^4 4!}, \quad P = \frac{(5!)^4 4!}{20!}$$

Q2:

$$\frac{1}{3}, \quad \frac{1}{3}$$

Q3:

$$0.907$$

Q4:

$$0.8187, \quad 0.0175$$

Q5:

$$0.8641, \quad 0.1359$$

Q6:

$$0.5768, \quad 0.607$$

Q7:

$$1 - Q\left(\frac{3}{2}\right),\; Q\left(\frac{7}{2}\right),\; 1 - 2Q(1),\; 1 - Q(2) + Q(3)$$

Q8:

Full conditional binomial expression as above.

Q9:

$$e^{-\lambda}, \quad 1 - e^{-\lambda} - \lambda e^{-\lambda} - \frac{\lambda^2 e^{-\lambda}}{2}$$

Q10:

$$p > \frac{1}{2}$$

End of Scribe.